

HPS Gaussian-Process Resonance Search

Signal-tail study v5.9
Gaussian cores with broader tails

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Standalone study following Analysis Note v5.0.5

Abstract

This study tests symmetric signal shapes with a Gaussian core inside the primary $\pm 2.25\sigma_m$ blind window and broader tails outside it. The tail scale is increased by 10, 20 and 30% in two specified shape families. Observed 90% CL_s limits and fixed-mass, one-sided asymptotic background-only probabilities are evaluated for full 2015 over 19–100 MeV, full 2016 over 39–180 MeV, and the released 2021 10% sample over 50–250 MeV. Each family is compared with a Gaussian under the same background and window prescription. The primary calculation retains the v5.0.5 window; a separate $\pm 4.5\sigma_m$ calculation tests the information carried by the outer structure. Detector resolution, prompt-density normalization and archived kernel coordinates are held fixed. The results retain primary-window maxima at 51, 90 and 78 MeV, respectively. The smooth-dilation alternatives increase the median primary limits by approximately 0.168, 0.345 and 0.529%, with nearly unchanged local probabilities. These are declared signal-shape scenarios, conditional on the stated choices. They do not provide coverage calibration or a new global discovery probability.

1 Observed comparisons and scope

The requested tail changes are implemented outside a fixed Gaussian core. They are changes in a tail-scale parameter, rather than a 10–30% increase in the detector resolution or in the total number of signal events. Each template is normalized to unit yield over the inherited support, and the fitted amplitude denotes that full-template yield.

There are seven shapes in each window: one Gaussian, three smooth tail dilations and three tail-curvature variations. The three datasets are analyzed separately. No dataset combination or choice among scenarios is used to strengthen an observed limit or significance. The complete mass-by-mass results accompany the note; the peak tables identify the smallest local probability within each stated scan, without assigning a probability to the selection of that minimum.

1.1 The primary-window results

Table 1 records the strongest local excess for every primary-window scenario. The Gaussian row is freshly computed with the same portable fitting machinery used for all alternatives. For each alternative, the coupling and yield limits refer to its own full-template normalization, while the local probability tests whether a positive signal component improves the fit at that mass.

The primary-window local maxima remain at 51 MeV (2015), 90 MeV (2016), and 78 MeV (2021 10%) for all six altered shapes. For the primary dilation model, the median observed limit increases by approximately 0.168%, 0.345%, and 0.529% for 10%, 20%, and 30% broader tails. These small shifts follow the change in core probability; they are not 10–30% changes in the full-template resolution.

In the guard window, the 30% dilation changes observed limits by -0.048 to $+2.068\%$ in 2015, $+0.245$ to $+2.056\%$ in 2016, and $+0.279$ to $+1.543\%$ in 2021. The largest absolute changes in the signed local root are 0.0377, 0.0280, and 0.0177, respectively. The guard-window Gaussian maximum in 2015 is already at 93 MeV, rather than 51 MeV. That movement is a consequence of the wider background interpolation and fitted region; it cannot be attributed to the tail variation. The primary search is complete over all declared mass intervals. The guard search omits 2016 masses 176–180 MeV because fewer than three upper exterior bins remain.

Dataset	Shape	m_0 (MeV)	Z_0	Local p_0	ε_{90}^2
2015	Gaussian	51	3.13924	8.469×10^{-4}	2.640×10^{-5}
2015	Dilation +10%	51	3.13925	8.469×10^{-4}	2.644×10^{-5}
2015	Dilation +20%	51	3.13926	8.469×10^{-4}	2.649×10^{-5}
2015	Dilation +30%	51	3.13927	8.469×10^{-4}	2.654×10^{-5}
2015	Curvature +10%	51	3.13924	8.469×10^{-4}	2.641×10^{-5}
2015	Curvature +20%	51	3.13924	8.469×10^{-4}	2.642×10^{-5}
2015	Curvature +30%	51	3.13924	8.469×10^{-4}	2.643×10^{-5}
2016	Gaussian	90	3.42475	3.077×10^{-4}	1.313×10^{-5}
2016	Dilation +10%	90	3.42475	3.077×10^{-4}	1.316×10^{-5}
2016	Dilation +20%	90	3.42475	3.077×10^{-4}	1.318×10^{-5}
2016	Dilation +30%	90	3.42475	3.077×10^{-4}	1.320×10^{-5}
2016	Curvature +10%	90	3.42475	3.077×10^{-4}	1.314×10^{-5}
2016	Curvature +20%	90	3.42475	3.077×10^{-4}	1.315×10^{-5}
2016	Curvature +30%	90	3.42475	3.077×10^{-4}	1.315×10^{-5}
2021	Gaussian	78	2.80864	2.488×10^{-3}	6.330×10^{-6}
2021	Dilation +10%	78	2.80864	2.488×10^{-3}	6.341×10^{-6}
2021	Dilation +20%	78	2.80864	2.488×10^{-3}	6.352×10^{-6}
2021	Dilation +30%	78	2.80864	2.488×10^{-3}	6.363×10^{-6}
2021	Curvature +10%	78	2.80864	2.488×10^{-3}	6.333×10^{-6}
2021	Curvature +20%	78	2.80864	2.488×10^{-3}	6.336×10^{-6}
2021	Curvature +30%	78	2.80864	2.488×10^{-3}	6.338×10^{-6}

Table 1: Primary-window scan minima and corresponding observed limits. Every probability is local, with the mass and shape fixed. A scan minimum does not include a mass or scenario trials correction.

2 Signal shapes and normalization

Let $z = |m - m_0|/\sigma_m$, $a = 2.25$ and $t = z - a$. The nominal resolution $\sigma_m = \sigma_m(m_0)$ is inherited from v5.0.5. All shapes have the unnormalized core

$$f(z) = \exp(-z^2/2), \quad 0 \leq z \leq a. \quad (1)$$

For $z > a$, the primary family stretches the tail coordinate smoothly:

$$u_k(z) = a + \frac{t}{1 + (k-1)[1 - \exp(-t/h)]}, \quad h = 0.5, \quad (2)$$

$$f_{\text{dil},k}(z) = \exp[-u_k(z)^2/2], \quad k \in \{1.1, 1.2, 1.3\}. \quad (3)$$

The coordinate and first derivative match the Gaussian at $z = a$; far into the tail, the distance from the boundary is stretched by k . The secondary family instead changes the tail curvature:

$$f_{\text{curv},k}(z) = \exp\left[-\frac{a^2}{2} - at - \frac{t^2}{2k^2}\right], \quad z > a. \quad (4)$$

It also matches the core value and slope at the boundary. The two families test different continuations of the same core; their common k does not imply equal tail fractions or equal root-mean-square widths. Both reduce exactly to the Gaussian at $k = 1$.

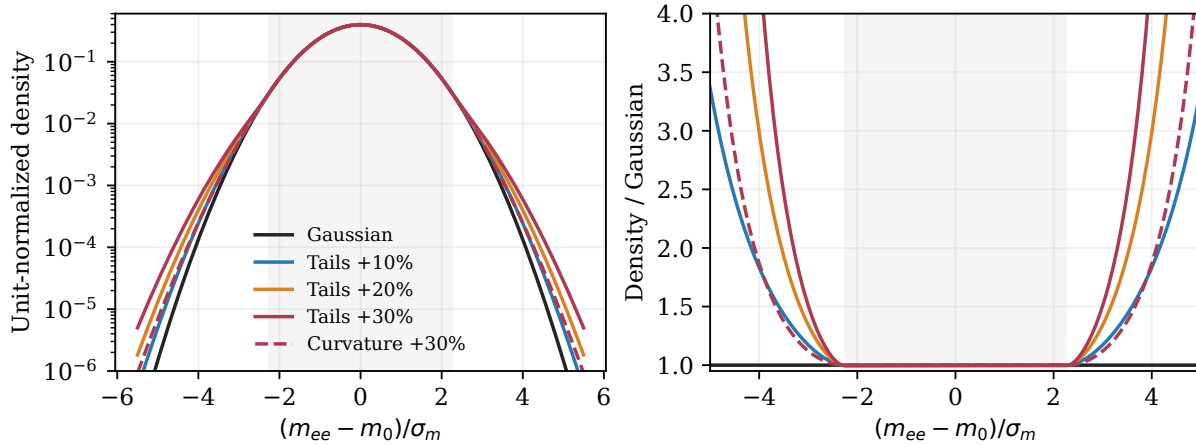


Figure 1: Smooth-dilation signal shapes and the 30% curvature control in units of the nominal mass resolution. The left panel shows full-line-normalized densities and the right shows their ratios to the Gaussian. Changes to the unnormalized shape begin at the primary blind-window boundary.

Core-only bins differ solely by normalization, leaving their unrestricted likelihood ratio and p_0 invariant. The actual center-selected bins can straddle $\pm 2.25\sigma_m$; their integrated tails break this proportionality and can produce small primary-probability changes. The guard calculation explicitly tests outer bins.

2.1 Bin integration and the full-template yield

For support $\mathcal{D}_d$ and analysis bin B_i in dataset d , the template probability is

$$T_{di}^{(s)}(m_0) = \frac{\int_{B_i} f_s(|m - m_0|/\sigma_m) dm}{\int_{\mathcal{D}_d} f_s(|m - m_0|/\sigma_m) dm}, \quad \sum_{i \in \mathcal{D}_d} T_{di}^{(s)} = 1. \quad (5)$$

The integrated bin probabilities are retained when a fit window is selected. They are never renormalized inside that window. If $F_{d,s,W} = \sum_{i \in W} T_{di}^{(s)}$, the fitted-window yield at the upper limit is $A_{90} F_{d,s,W}$, while A_{90} itself denotes the full-template yield. The normalization in Eq. (5) also accounts for the finite support near its endpoints.

Shape	Tail (%)	RMS/ σ_m	Core UL ratio	Beyond guard
Gaussian	2.4449	1.00000	1.000000	6.795×10^{-6}
Dilation +10%	2.6084	1.00735	1.001679	1.902×10^{-5}
Dilation +20%	2.7798	1.01538	1.003445	4.387×10^{-5}
Dilation +30%	2.9583	1.02406	1.005290	8.773×10^{-5}
Curvature +10%	2.4926	1.00225	1.000490	1.154×10^{-5}
Curvature +20%	2.5320	1.00416	1.000894	1.739×10^{-5}
Curvature +30%	2.5649	1.00578	1.001231	2.403×10^{-5}

Table 2: Numerical shape properties for the declared tail prescriptions, evaluated on the full line. The core upper-limit ratio is the ideal proportional-core normalization ratio. Actual fitted templates use the finite support in Eq. (5).

The prompt-density normalization uses the same nominal resolution for every shape:

$$\rho_d(m_0) = \frac{1}{3.28\sigma_m} \int_{m_0 - 1.64\sigma_m}^{m_0 + 1.64\sigma_m} \frac{dN_d}{dm} dm, \quad K_d(m_0) = \frac{3\pi m_0 f_{\text{rad},d}^{\text{eff}}}{2\alpha} \rho_d(m_0). \quad (6)$$

The integral uses fractional overlaps with the native histogram bins. The electron-channel coupling coordinate satisfies $A = K_d \varepsilon^2$. Holding K_d fixed isolates the specified shape change; this study does not infer a new efficiency or acceptance from the broader tails.

Above the dimuon threshold, the inherited minimal-visible coupling correction is applied once:

$$(\varepsilon_{90}^2)_{\text{minimal}} = \frac{A_{90}}{K_d} c_\mu(m_0), \quad c_\mu(m_0) = \begin{cases} 1, & m_0 \leq 2m_\mu, \\ 1 + \sqrt{1 - 4r_\mu}(1 + 2r_\mu), & m_0 > 2m_\mu, \end{cases} \quad r_\mu = (m_\mu/m_0)^2. \quad (7)$$

This conversion changes neither the visible signal-yield limit nor p_0 . In the numerical ledger, **A90** is the solver coordinate $\varepsilon^2/(10^{-8})$, whereas **full_yield_90** is the event yield denoted A_{90} in this note; **derived/COLUMN_GUIDE.md** defines the columns and conversions.

3 Primary-window observed scans

3.1 Full 2015: 19–100 MeV

The 2015 scan retains the 14–135 MeV support and 0.25 MeV analysis bins. The 91–100 MeV extension follows the v5.0.5 endpoint prescription: kernel coordinates are fixed at their archived 90 MeV values, while the resolution, mask, template and density are evaluated at the actual mass. The extrapolated interval keeps its conditional status.

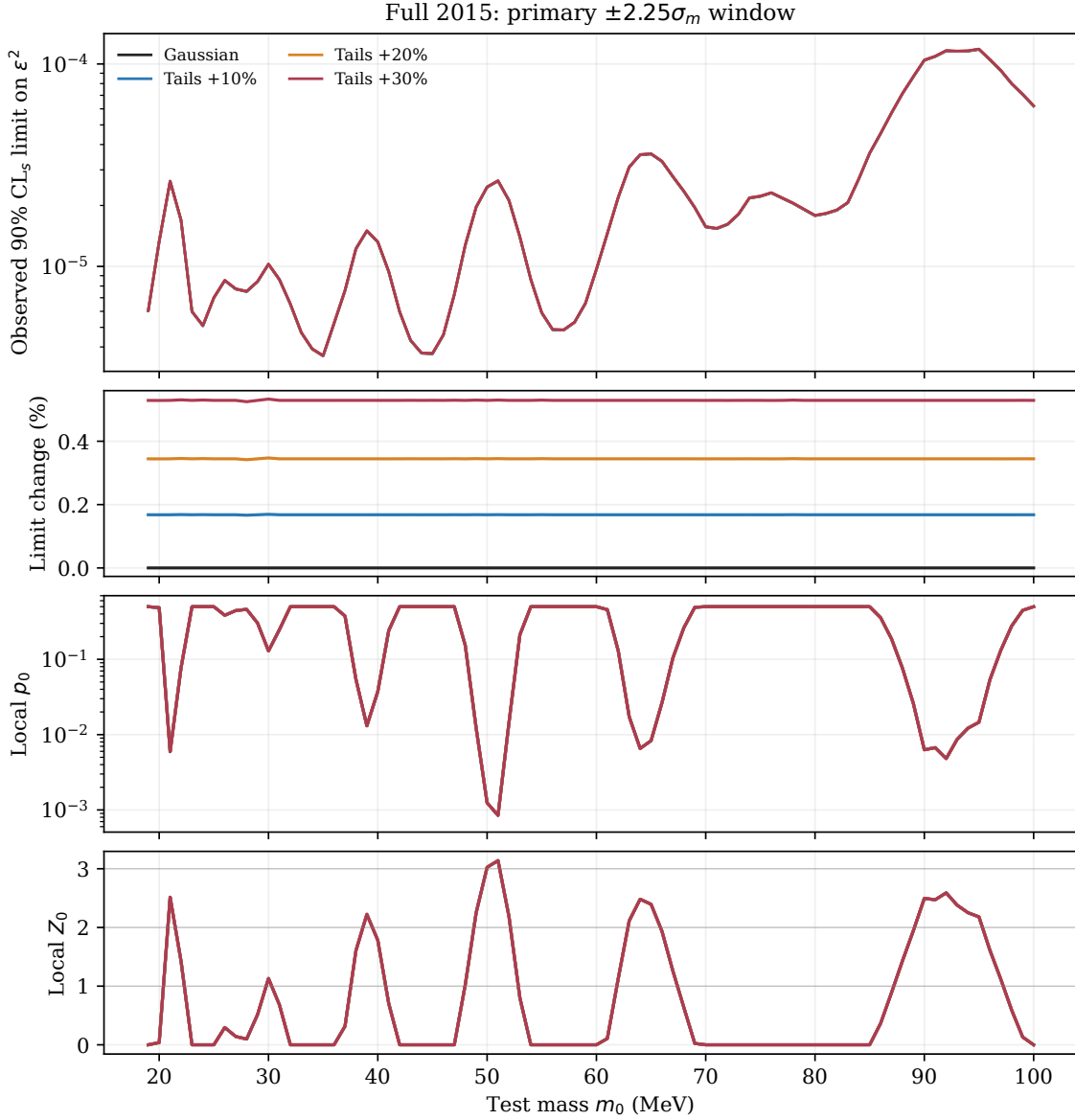


Figure 2: Observed 2015 primary-window Gaussian and smooth-dilation scenarios. The coupling conversion refers to the complete normalized template; the likelihood uses the selected $\pm 2.25\sigma_m$ bins. Limits are pointwise asymptotic 90% CL_s limits and probabilities are fixed-mass local references.

3.2 Full 2016: 39–180 MeV

The 2016 scan uses the frozen 30–210 MeV support and 0.25 MeV analysis bins. Every shape at a mass uses the same archived kernel coordinates and the same background constraint for this window. Thus the within-window comparison changes the signal template without selecting a new background fit from the resulting limit or probability.

The v5.0.5 note documents differences in independent 2016 hyperparameter fits. Reusing its archived states makes this calculation reproducible under that prescription; it does not settle the separate fit-provenance or background-calibration questions.

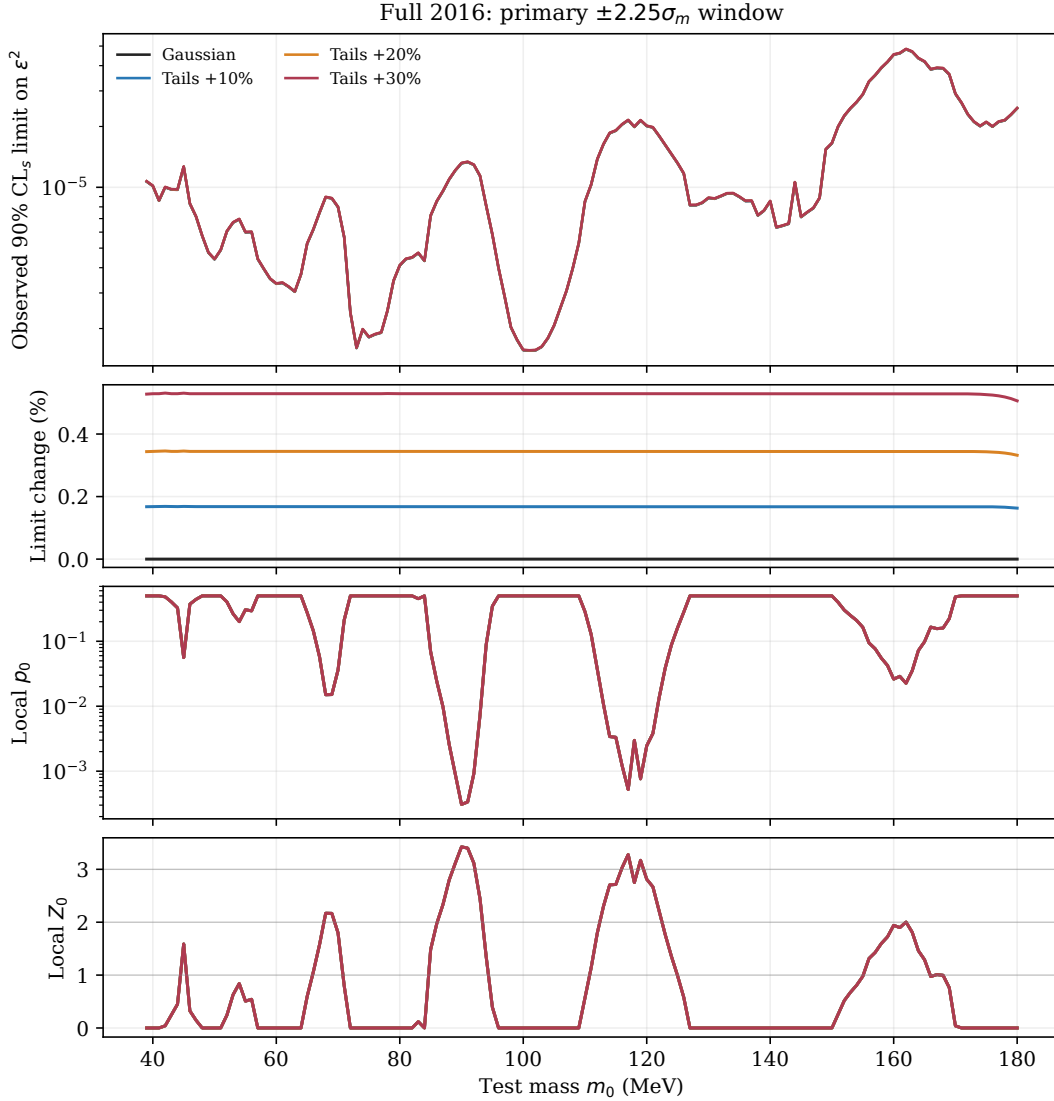


Figure 3: Observed 2016 primary-window Gaussian and smooth-dilation results. The same frozen-state qualification applies to the Gaussian and every tail alternative. Probability differences should be interpreted together with the small boundary-bin contributions and the explicitly broader calculation below.

3.3 Released 2021 10%: 50–250 MeV

The 2021 scan uses the native 10% sample with the inherited 36–300 MeV support prescription. Rebinning leaves the final complete analysis-bin edge at 299.75 MeV; the analysis-bin width is 0.625 MeV. The larger bin width makes the distinction between a continuous mask boundary and a full selected bin particularly relevant to the core-only comparison.

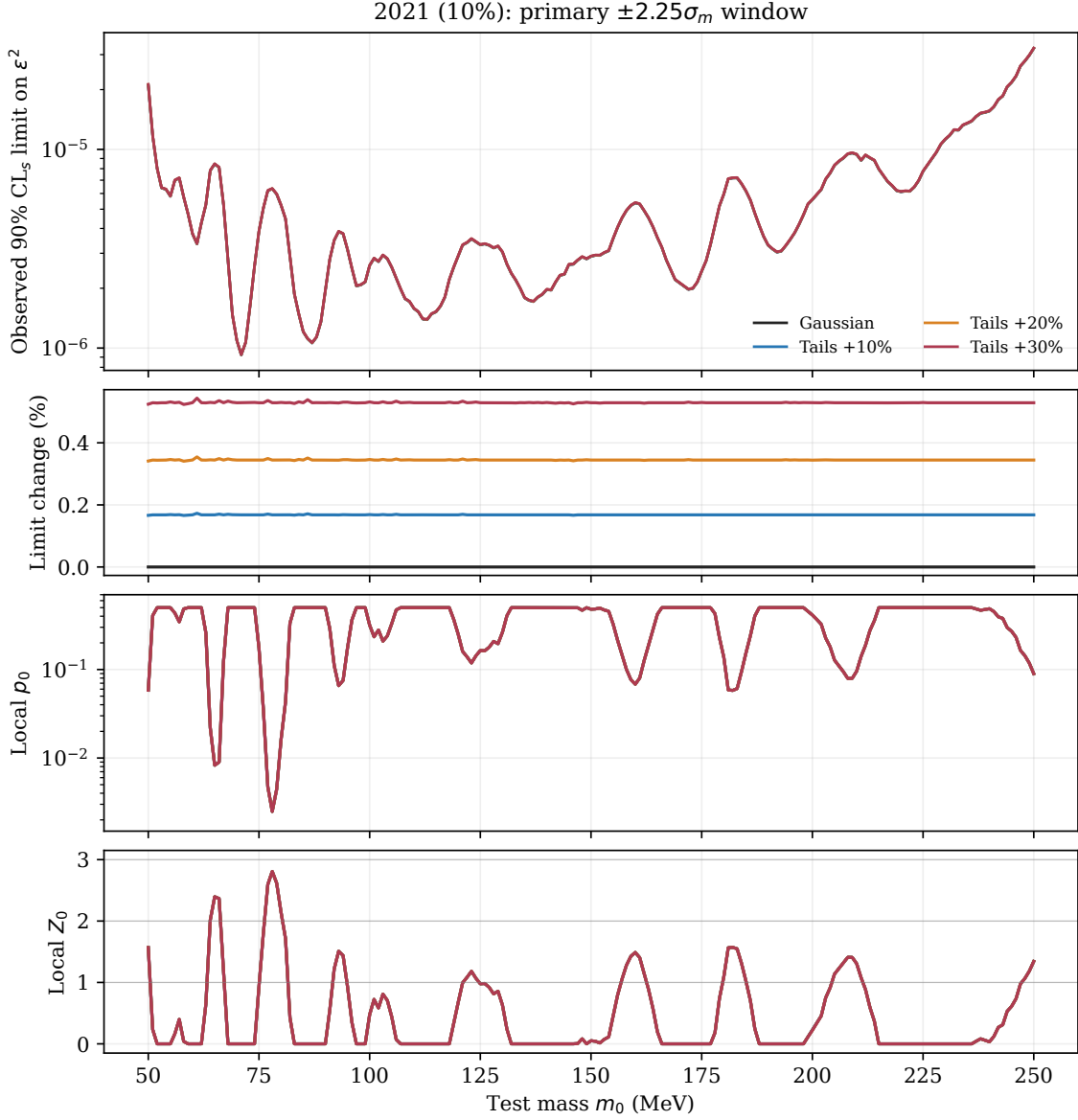


Figure 4: Observed 2021 10% primary-window Gaussian and smooth-dilation results. Above $2m_\mu$, the plotted coupling includes the inherited branching correction once. Neither the visible yield nor the local probability receives this correction.

4 Outer structure in a wider fit region

The guard calculation expands both the fitted region and the training exclusion to $\pm 4.5\sigma_m$. Its GP is conditioned only on bins outside that wider mask, using the same archived kernel coordinates. The prediction and covariance are recomputed for the widened region. Every observed bin in that region enters the likelihood once.

The primary scans contain all 425 requested dataset–mass coordinates. The guard scan contains 420: the five 2016 hypotheses at 176–180 MeV do not retain the required external sidebands under the wider mask and are excluded by the geometric rule. Its 2016 domain is consequently 39–175 MeV. The 2015 and 2021 guard domains retain their full primary ranges. Across seven shapes and both windows, 5,915 observed fits are reported; omitted guard endpoints are not interpolated or filled from the primary calculation.

This is a separate background-interpolation and extraction prescription. The meaningful shape comparison is therefore the tail scenario divided by, or compared with, the Gaussian *in the same guard window*. Comparing a guard tail fit directly with a primary Gaussian would mix the tail change with the window and background change. The wider window has not undergone an independent support or coverage qualification.

Dataset	Shape	m_0 (MeV)	Z_0	Local p_0	ϵ_{90}^2
2015	Gaussian	93	3.52179	2.143×10^{-4}	1.453×10^{-4}
2015	Dilation +10%	93	3.51668	2.185×10^{-4}	1.457×10^{-4}
2015	Dilation +20%	93	3.51147	2.228×10^{-4}	1.461×10^{-4}
2015	Dilation +30%	93	3.50616	2.273×10^{-4}	1.465×10^{-4}
2015	Curvature +10%	93	3.52029	2.155×10^{-4}	1.454×10^{-4}
2015	Curvature +20%	93	3.51907	2.165×10^{-4}	1.455×10^{-4}
2015	Curvature +30%	93	3.51806	2.174×10^{-4}	1.456×10^{-4}
2016	Gaussian	91	3.46714	2.630×10^{-4}	1.388×10^{-5}
2016	Dilation +10%	91	3.46647	2.637×10^{-4}	1.392×10^{-5}
2016	Dilation +20%	91	3.46582	2.643×10^{-4}	1.397×10^{-5}
2016	Dilation +30%	91	3.46513	2.650×10^{-4}	1.401×10^{-5}
2016	Curvature +10%	91	3.46701	2.631×10^{-4}	1.389×10^{-5}
2016	Curvature +20%	91	3.46691	2.632×10^{-4}	1.390×10^{-5}
2016	Curvature +30%	91	3.46682	2.633×10^{-4}	1.391×10^{-5}
2021	Gaussian	78	2.76302	2.863×10^{-3}	6.469×10^{-6}
2021	Dilation +10%	78	2.76074	2.884×10^{-3}	6.486×10^{-6}
2021	Dilation +20%	78	2.75828	2.905×10^{-3}	6.504×10^{-6}
2021	Dilation +30%	78	2.75564	2.929×10^{-3}	6.521×10^{-6}
2021	Curvature +10%	78	2.76235	2.869×10^{-3}	6.474×10^{-6}
2021	Curvature +20%	78	2.76180	2.874×10^{-3}	6.478×10^{-6}
2021	Curvature +30%	78	2.76132	2.878×10^{-3}	6.481×10^{-6}

Table 3: Guard-window scan minima and their corresponding observed limits. These rows are separate fixed-shape local calculations, with a Gaussian guard-window reference.

4.1 Guard-window 2015 comparison

Widening the excluded region removes sideband information and adds explicit tail-sensitive bins to the likelihood. Both effects are included in the Gaussian guard reference. Near a support edge, the available external sidebands need not be symmetric; the fixed support and actual bin selection remain part of the definition.

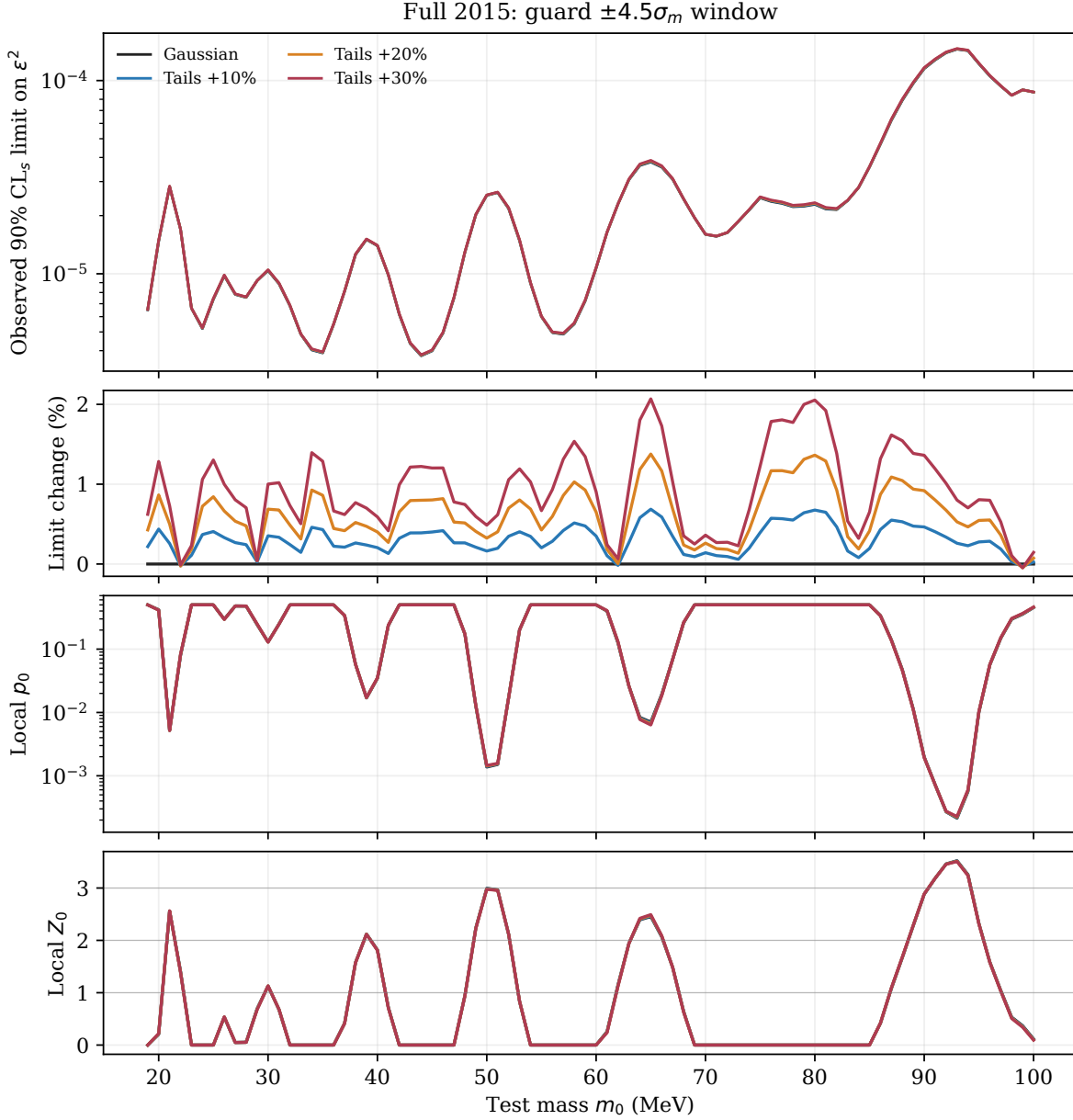


Figure 5: Observed 2015 $\pm 4.5\sigma_m$ fits with external GP training. The 91–100 MeV endpoint prescription is retained. Tail effects are assessed relative to the Gaussian fitted with this same wider mask.

4.2 Guard-window 2016 comparison

Broader fitted structure can overlap background directions permitted by the correlated GP constraint. The profiled likelihood accounts for that overlap through its nuisance parameters. An observed change in p_0 or the upper limit consequently measures the response of the complete specified fit, rather than a direct measurement of the detector's signal tail.

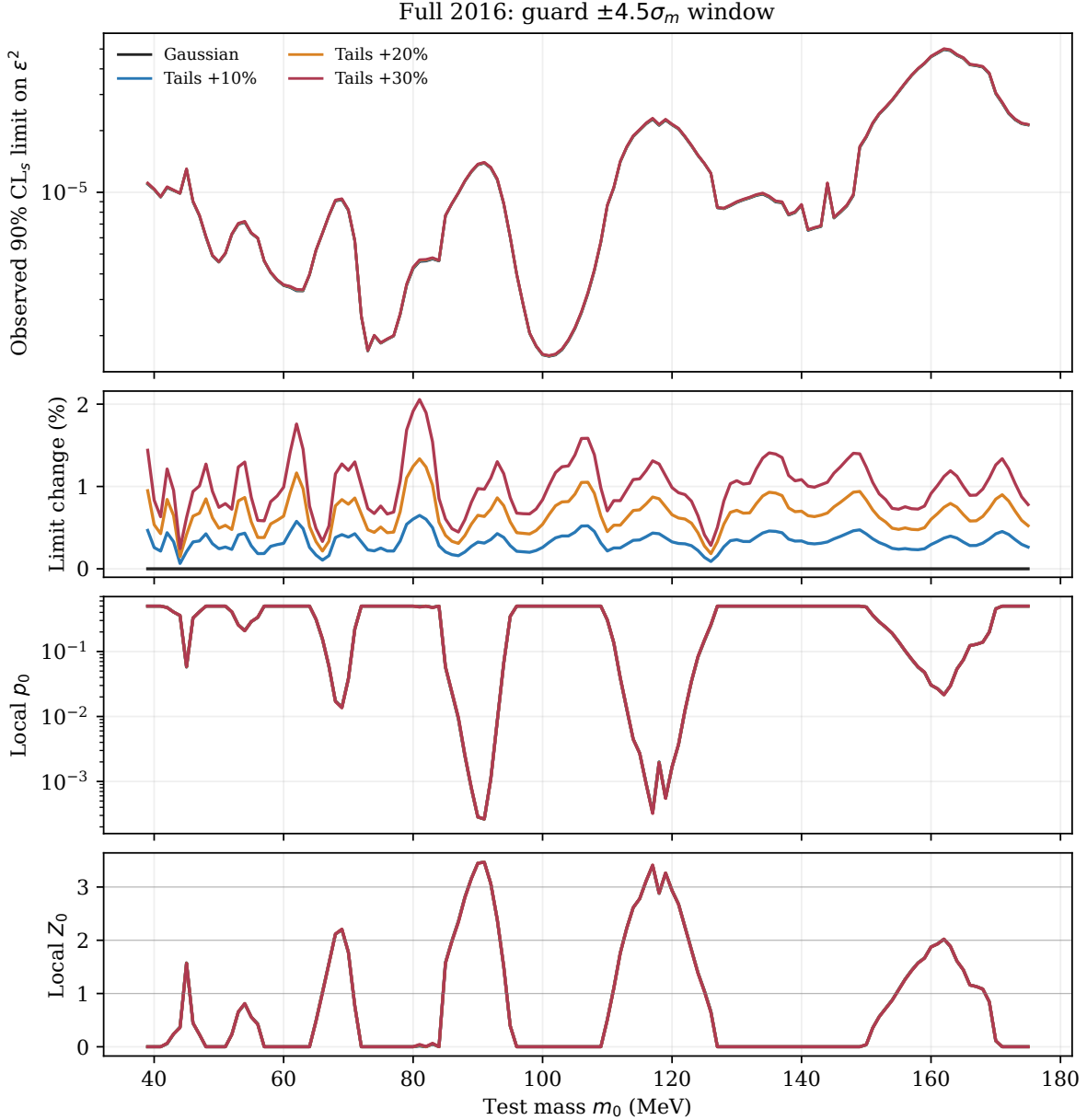


Figure 6: Observed 2016 $\pm 4.5\sigma_m$ fits over 39–175 MeV. The final five primary-scan masses fail the external-sideband geometric requirement for this wider mask. The GP mean and covariance are held fixed across signal hypotheses within each mass and window; nuisance shifts are profiled at every tested amplitude. The archived 2016 state qualification remains in force.

4.3 Guard-window 2021 comparison

The 2021 guard scan retains the original 10% exposure and normalization. It is not an exposure projection. Its broader likelihood region tests whether the declared outer template changes the pointwise inference when those bins are used explicitly.

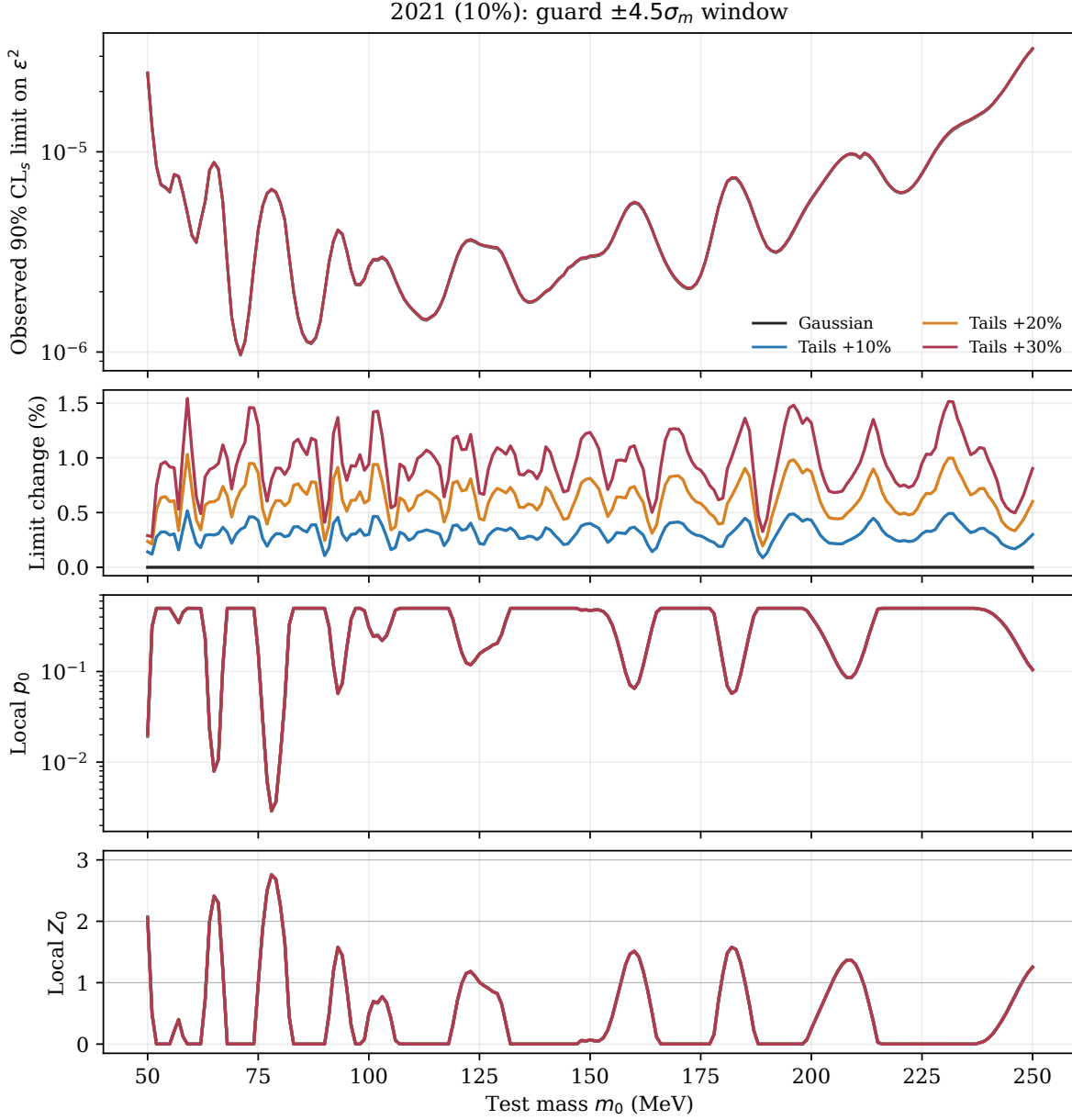


Figure 7: Observed 2021 10% $\pm 4.5\sigma_m$ fits. Each alternative is compared with the Gaussian under the same support, external training set and fit window.

5 Size and location of the response

The ratios below compare each alternative with the Gaussian from the same dataset, mass and window. They quantify changes in the reported observed limit, not expected sensitivity. The collection of scenarios is not a confidence band or a nuisance-profiled uncertainty interval.

Dataset	Shape	Primary median (%)	Guard range (%)	Max. $ \Delta r $ (guard)
2015	Dilation +10%	0.168	-0.030–0.687	0.01276
2015	Dilation +20%	0.345	-0.047–1.378	0.02537
2015	Dilation +30%	0.529	-0.048–2.068	0.03773
2015	Curvature +10%	0.049	-0.007–0.195	0.00358
2015	Curvature +20%	0.089	-0.010–0.352	0.00642
2015	Curvature +30%	0.123	-0.015–0.480	0.00871
2016	Dilation +10%	0.168	0.065–0.649	0.00866
2016	Dilation +20%	0.345	0.147–1.336	0.01804
2016	Dilation +30%	0.529	0.245–2.056	0.02796
2016	Curvature +10%	0.049	0.025–0.192	0.00268
2016	Curvature +20%	0.089	0.048–0.351	0.00491
2016	Curvature +30%	0.123	0.068–0.483	0.00678
2021	Dilation +10%	0.168	0.087–0.517	0.00623
2021	Dilation +20%	0.345	0.196–1.032	0.01215
2021	Dilation +30%	0.529	0.279–1.543	0.01772
2021	Curvature +10%	0.049	0.023–0.143	0.00167
2021	Curvature +20%	0.089	0.036–0.257	0.00310
2021	Curvature +30%	0.123	0.042–0.349	0.00439

Table 4: Observed percent limit changes relative to the Gaussian in the same window. Positive values denote larger upper limits.

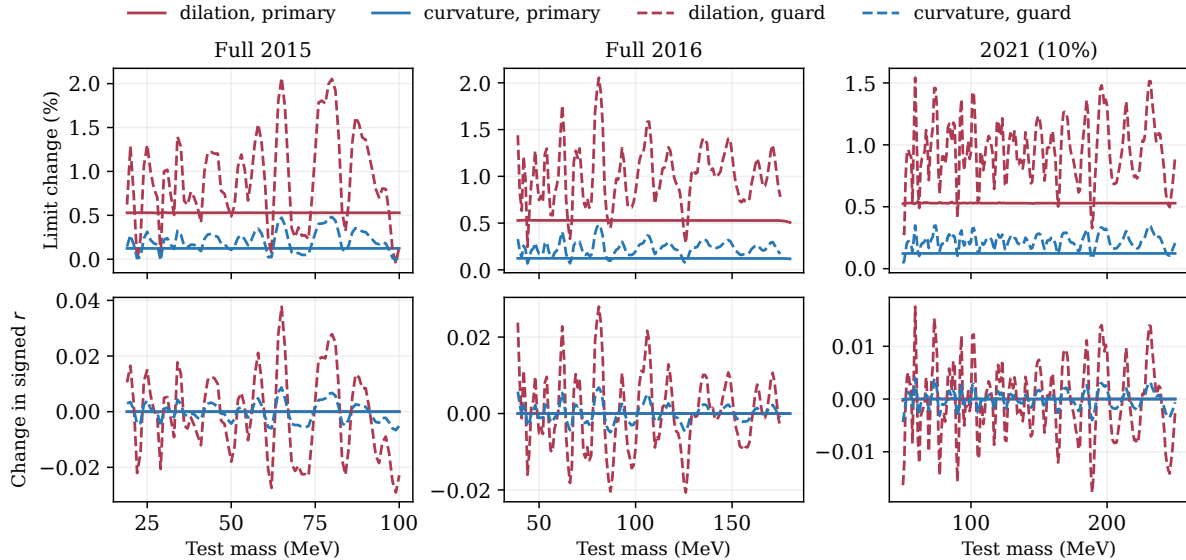


Figure 8: Response of the observed inference at the largest declared tail change, $k = 1.3$: percent changes in the limit and changes in the signed likelihood root are compared with the Gaussian at each mass. Both tail families are shown, and primary and guard results retain their separate Gaussian references. Features in a mass scan reuse sideband information and should not be counted as independent evidence.

6 Signal in the GP training region

Signal tails can enter bins used for background-only GP training. Changing the local signal template does not correct that contamination. A wider mask can reduce the training contribution, while also removing background information.

The diagnostic uses anchors 51 and 92 MeV in 2015, 90 and 92 MeV in 2016, and 78 and 92 MeV in 2021. At each anchor the full injected yield is fixed to the Gaussian primary observed A_{90} , retaining the nominal conversion. Each window has its own smooth source n_0 , the full-support GP prediction from the observed external bins. Reconditioning on the external bins of n_0 gives b_0, C_0 . The deterministic fitted-bin expectation is then $n = b_0 + A_{\text{true}}T_{\text{truth}}$.

The clean-training branch fits this expectation with b_0, C_0 . The contaminated-training branch adds the same full signal to n_0 before reconditioning, obtains b_1, C_1 , and fits exactly the same fitted-bin expectation. Both branches are evaluated with the matching signal template and with a Gaussian extraction template. Comparing $\hat{A}/A_{\text{true}}$ isolates the declared training response and template mismatch. Clean matched recovery equals one by construction: centering on b_0 makes this a numerical identity, not an independent background or injection-closure test. Primary and guard sources are constructed separately, so their difference also changes the reference continuum; it is not a comparison of windows under one common truth.

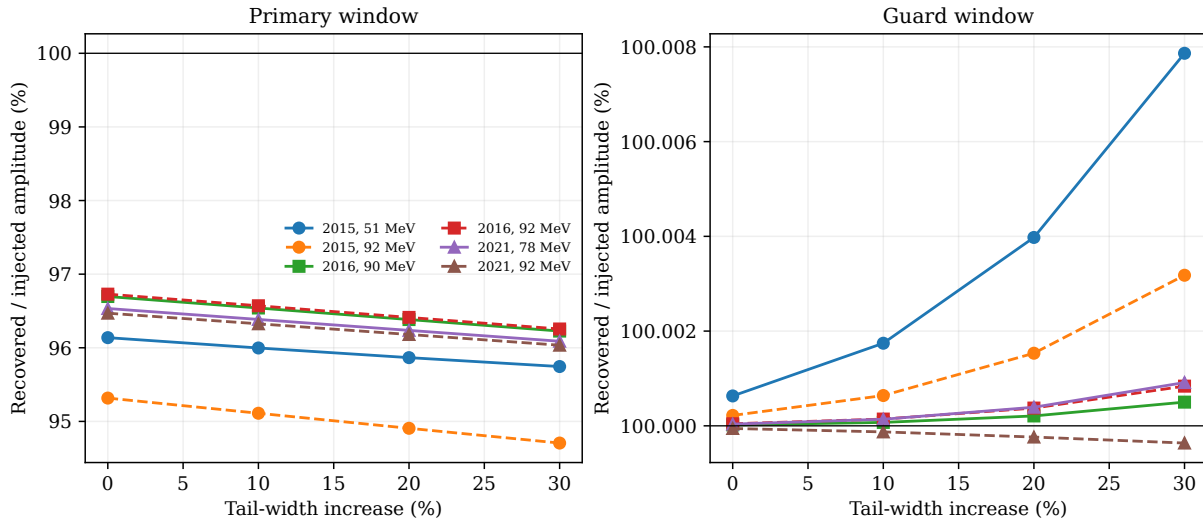


Figure 9: Deterministic signal-leakage diagnostic: recovered yield divided by injected yield for contaminated training with matched Gaussian or smooth-dilation extraction. The horizontal line marks unit recovery. Clean-training, Gaussian-mismatch and curvature controls are retained in the numerical ledger. The injected yield is the Gaussian primary observed A_{90} at each anchor.

No random sampling or coverage test is performed. These conditional responses do not establish that an observed structure arose from signal leakage or that an alternative template is the true detector response.

7 Likelihood and probability construction

For one dataset, mass, window and shape, the observed expectation is

$$\lambda_i = b_i + (L\boldsymbol{\theta})_i + AT_i, \quad LL^T = C, \quad (8)$$

with profiled negative log likelihood, up to a data-only constant,

$$\ell(A, \boldsymbol{\theta}) = \sum_i \left[\lambda_i - n_i + n_i \log \frac{n_i}{\lambda_i} \right] + \frac{1}{2} \boldsymbol{\theta}^T \boldsymbol{\theta}. \quad (9)$$

The zero-count contribution is λ_i . The GP count mean b and covariance C are determined by the external sidebands. Poisson expectations are constrained to remain physical. The same b, C, T enter the null, unrestricted and tested-amplitude fits at a fixed coordinate. The unrestricted amplitude may be negative, enabling a signed diagnostic; the physical signal strength for limits is nonnegative.

Writing $\hat{A}$ for the unrestricted best fit, define

$$r = \text{sign}(\hat{A}) \sqrt{2[\ell(0, \hat{\boldsymbol{\theta}}_0) - \ell(\hat{A}, \hat{\boldsymbol{\theta}})]}, \quad (10)$$

$$Z_0 = \max(r, 0), \quad p_0 = \Phi(-Z_0). \quad (11)$$

Thus a deficit has $Z_0 = 0$ and $p_0 = 0.5$ by the inherited one-sided reference convention. This is distinct from the signed diagnostic $p_{\text{signed}} = \Phi(-r)$.

For a tested $A \geq 0$, the bounded upper-limit statistic is zero if $A < \hat{A}$. Otherwise its denominator uses the unrestricted fit when $\hat{A} \geq 0$ and the null fit when $\hat{A} < 0$. Let q be that observed statistic and q_A the explicitly profiled background-Asimov statistic at the same tested amplitude. With $a_A = \sqrt{q_A}$, the asymptotic tails are

$$(z_{s+b}, z_b) = \begin{cases} (\sqrt{q}, \sqrt{q} - a_A), & q \leq q_A, \\ \left(\frac{q + q_A}{2a_A}, \frac{q - q_A}{2a_A} \right), & q > q_A, \end{cases} \quad CL_s(A) = \frac{\Phi(-z_{s+b})}{\Phi(-z_b)}. \quad (12)$$

The $q = 0$ inclusive-tail convention gives $CL_s = 1$. The observed upper limit solves $CL_s(A_{90}) = 0.10$. Both likelihood ratios are profiled explicitly; their sampling tails use the asymptotic approximation [2]. No Gaussian error-bar substitution is used for the observed upper limit.

8 Input provenance, checks and interpretation

The portable spectra and fitting machinery come from Analysis Note v5.0.5 [1]. Native counts, bin edges, resolution, normalization and archived GP states are preserved, with original ROOT identities in the export provenance and exact copied-file hashes in the study manifest.

The Gaussian comparison is recomputed rather than copied from the historical results table. Its primary peak significances are 3.139237, 3.424752 and 2.808645 for 2015, 2016 and 2021, compared with the published-note values 3.139465, 3.423687 and 2.810290. Across the full primary scans the largest difference from the inherited table is $|\Delta Z| = 0.12366$, at 83 MeV in 2016; the largest relative limit difference is about 1.14%. At 2016/83 MeV the legacy L-BFGS null fit stopped on relative objective change with maximum gradient about 0.116; its $Z_0 = 0.24369$ becomes 0.12003 in the converged fit. Holding the background, covariance and template fixed reproduces this difference; the covariance-factor choice changes the converged limit by less than 6×10^{-12} relative. This numerical issue is distinct from the 2016 kernel-fit caveat. All tail comparisons use the newly converged Gaussian with identical settings.

The RBF kernel acts in log mass with the inherited log-count noise prescription. Its coordinates remain fixed across shapes. Effective radiative fractions are 0.07905, 0.04650 and 0.04770 for the three datasets, including the inherited penalties. No setting is selected by an observed limit or probability.

Quantity	Value
Observed mass–window–shape fits	5915
Primary / guard mass coordinates	425 / 420
Maximum $ \text{CL}_s - 0.1 $	1.382×10^{-8}
Maximum profile score residual	1.999×10^{-7}
Minimum fitted Poisson mean	62.458
Maximum historical-ledger relative UL difference	1.139%
Maximum historical-ledger $ \Delta Z_0 $	0.123661
Deterministic response fits	336
Maximum clean matched recovery error	1.317×10^{-8}

Table 5: Recorded numerical and artifact checks. Passing these checks establishes the stated numerical calculation; it does not establish frequentist coverage or the physical correctness of a chosen tail model.

Choosing a preferred shape requires an independent detector-response model or a calibrated selection procedure. These scenario comparisons supply no new expected bands, coherent scan calibration, look-elsewhere correction or global probability.

References

- [1] E. Peets, M. Gignac and E. Hsu, *HPS Gaussian-Process Resonance Search: Analysis Note v5.0.5*, 16 September 2026, internal review draft and archived source package.
- [2] G. Cowan, K. Cranmer, E. Gross and O. Vitells, “Asymptotic formulae for likelihood-based tests of new physics,” *Eur. Phys. J. C* **71**, 1554 (2011), arXiv:1007.1727.